Yan Liang

51 Pegasi b Postdoctoral Fellow, Yale University
https://orcid.org/0000-0002-1001-1235; yan.liang@yale.edu

Academic Appointments

Yale University ***Aug 2025 – Present***
51 Pegasi b Postdoctoral Fellow, Department of Astronomy
Faculty Host: Malena Rice

Education

Princeton University ***Sep 2020 – May 2025***
Ph.D., Department of Astrophysical Sciences *GPA: 3.90/4.00*
Advisor: Joshua N. Winn, **Co-Advisor:** Peter Melchior

University of California, Berkeley ***Aug 2018 – Dec 2019***
Physics BA, College of Letters and Science *GPA: 4.00/4.00*
Advisor: Uros Seljak

University of California, San Diego ***Sep 2016 – Jun 2018***
Undergraduate study in Physics, Astrophysics Specialization *GPA: 3.99/4.00*
Advisor: Thomas Murphy

Publications

AESTRA II: Generative Spectral Modeling of the Sun as a Star for Precise Radial Velocities
Revised Manuscript Submitted to AAS Journals October 2026, under review.
Yan Liang; Joshua N. Winn; Peter Melchior; Sicong Lu; Quang Tran

Benchmarking Machine Learning Emulators of Stellar Evolution for Precision Asteroseismology
The Astrophysical Journal (2026), Volume 1010, Issue 1, id.33
Naomi Gluck, Earl P. Bellinger, **Yan Liang**, Ebraheem Farag, Nicholas Saunders, Selim Kalici, Christopher J. Lindsay, and Sarbani Basu

AESTRA: Deep Learning for Precise Radial Velocity Estimation in the Presence of Stellar Activity
The Astronomical Journal (2024), Volume 167, Issue 1, id.23
Yan Liang; Joshua N. Winn; Peter Melchior

Outlier Detection in the DESI Bright Galaxy Survey
The Astrophysical Journal Letters (2023), Volume 956, Issue 1, id.L6
Yan Liang; Peter Melchior; ChangHoon Hahn; Jeff Shen; Andy Goulding; Charlotte Ward

Autoencoding Galaxy Spectra. II. Redshift Invariance and Outlier Detection
The Astronomical Journal (2023), Volume 166, Issue 2, id.75
Yan Liang; Peter Melchior; Sicong Lu; Andy Goulding; Charlotte Ward

Autoencoding Galaxy Spectra. I. Architecture
The Astronomical Journal (2023), Volume 166, Issue 2, id.74
Peter Melchior; **Yan Liang**; ChangHoon Hahn; Andy Goulding

DI Herculis Revisited: Starspots, Gravity Darkening, and 3D Obliquities
The Astrophysical Journal (2022), Volume 927, Issue 1, id.114
Yan Liang; Joshua N. Winn; Simon H. Albrecht

Kepler-90: Giant Transit-Timing Variations Reveal a Super-Puff
The Astronomical Journal (2021), Volume 161 , Issue 4, id.202
Yan Liang; Jakob Robnik; Uroš Seljak

APOLLO Clock Performance and Normal Point Corrections
Classical and Quantum Gravity (2017), Volume 34, Issue 24, article id. 245009
Yan Liang; Thomas Murphy; Nicholas Colmenares; James Battat

Awards and Honors

51 Pegasi b Postdoctoral Fellowship ***Feb 2025***
Awarded by the Heising-Simons Foundation for postdoctoral research in planetary astronomy at Yale University; provides up to $450,000 in support over three years.

Charlotte Elizabeth Procter Fellowship ***Aug 2024 – Jul 2025***
Awarded by Princeton Graduate School; includes tuition, fees, and a $56,760 stipend.

Highest Distinction In General Scholarship ***Dec 2019***
Awarded by the College of Letters and Science at the University of California, Berkeley.

Professional Service

NASA Exoplanets Research Program Grant Peer Reviewer ***Mar 2026***
Reviewed research proposals for NASA's Exoplanets Research Program.

Scientific Organizing Committee ***Jan 2026***
A Dynamical Synthesis of Planetary Systems, Yale University.

Selected Talks and Seminars

Generative AI for Doppler Detection of Earth-Like Exoplanets
Invited Talk, Harvard–Smithsonian Center for Astrophysics, Cambridge, MA Oct 2025

Disentangling Stellar Activity from Planet Signals with Generative Models
Yale Data Science × Astronomy–Astrophysics Seminar, New Haven, CT Mar 2026

Generative Spectral Modeling for Precise Radial Velocities
Invited Talk, THE Science Team Meeting, virtual Jun 2026

AESTRA: Deep Learning for Precise RV Estimation in the Presence of Stellar Activity
MIT Stellar Contamination Workshop, Cambridge, MA Sep 2024
The Exoplanets 5 Conference, Leiden, the Netherlands Jun 2024
EPRV & Machine Learning Workshop Jun 2024
THE Late Fall Science Meeting, University of Exeter, United Kingdom Nov 2023
ML Splinter Session, The EPRV 5 Conference, Santa Barbara, CA Mar 2023

Autoencoding Galaxy Spectra: Redshift Invariance and Outlier Detection
University Observatory Munich, Germany May 2023

Conference Posters

AESTRA II: Generative Spectral Modeling of the Sun as a Star for Precise Radial Velocities
International Conference on Exoplanets and Planet Formation, Shanghai, China Dec 2025

Outlier Detection in the DESI Bright Galaxy Survey
The Rare Gems in Big Data conference, Tucson May 2024

From Galaxy Outliers to Stellar Jitters: Deep Learning Tools for Next Generation Astronomy
243rd Meeting of The American Astronomical Society, New Orleans Jan 2024

Interpretable Encoding of Galaxy Spectra
Machine Learning and the Physical Sciences workshop, NeurIPS 2022 Dec 2022

Autoencoding Galaxy Spectra
ICML Workshop on Machine Learning for Astrophysics Jul 2022

Teaching Experience

Teaching Assistant, AST 204: Topics in Modern Astronomy ***Jan–May 2022***
Department of Astrophysics, Princeton University